



NGUYEN THANH DAT

Backend Engineer · NestJS / TypeScript · PHP / Laravel

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PROFESSIONAL SUMMARY

- **Nearly 5 years** of backend engineering across ERP, MES, healthcare, e-commerce and booking systems, mostly for Japanese clients — **PHP/Laravel throughout**, and for the past year **sole owner of a NestJS/TypeScript ERP service end to end**.
- **The seam is the specialty:** I keep a Laravel estate and a NestJS service agreeing in production. B2B and MES (PHP) write each day's business data into a shared sync database; **12 scheduled jobs and 3 RabbitMQ consumers** pull it across and recompute the ERP summaries.
- **Performance work is root-cause, not guesswork:** every result below traces to a query plan, a missing index or a composite index with the wrong column order, measured before and after.
- **Architecture and ownership:** set the architecture and technology stack from inception for a **5-person backend team**, and today I am the sole developer responsible for one production ERP service, which aggregates the B2B and MES systems beside it.

TECHNICAL SKILLS

Languages & Frameworks: PHP/Laravel (5 yrs) · TypeScript / JavaScript (2 yrs) · NestJS (1 yr) · Vue 3 (2 yrs) · Lumen

Databases & Performance: MySQL (3 yrs) · PostgreSQL (2.5 yrs) · TypeORM · Eloquent · Schema Design · Indexing · Query & Index Tuning · Execution-Plan Analysis · Redis Caching Strategies

Architecture & Async: Modular Architecture · RESTful API Design · Queue-Based / Async Processing (RabbitMQ, AWS SQS) · Scheduled Batch Jobs · Cross-System Data Sync · OAuth2 / JWT (Passport) · SSO (Keycloak)

Infrastructure & Deployment: Docker · Nginx · Container Resource Isolation · Linux · AWS S3 · GitHub Actions (CI/CD) · Cloudflare Pages · Git · GitHub · GitLab · Jira · Redmine · Backlog · Trello

Testing, Monitoring & Docs: Vitest · Unit & Feature Testing · Swagger · Laravel Telescope · Technical Documentation

Integrations: Stripe API · OpenAI API · Firebase Cloud Messaging · Pusher · Laravel Reverb · Google Maps API · SNS Login · Fax API

Working knowledge (side project): Gin · FastAPI · Caddy · Bun

PROFESSIONAL EXPERIENCE

BEAR ACER COMPANY — Backend Developer

Aug 2025 – Present | Japanese client | ERP service owned solo

- **Sole developer on the ERP service** — NestJS with TypeORM on the server, Vue on the client. Inherited the codebase and now own it end to end: architecture, features, performance and releases. It aggregates the B2B and MES PHP systems for daily/monthly reporting, customer support and email handling.
- **The client-facing monthly report:** 26s → 250ms. Replaced an on-the-fly aggregation over five million rows with a pre-computed rollup table, and rebuilt a composite index whose column order was wrong — **rows scanned 5M → 380K**.
- **Three CRM endpoints:** 18.2s → 64ms, 8.8s → 6ms, 6.1s → 187ms. The worst was a LEFT JOIN across ~18 tables (contacts 1M, contact info 2M rows); I split it into a fast main-record call plus a deferred contract call for the heavier, less urgent data.
- **A multi-column filter:** 9s → 80ms. The filter combination had no index to land on. Diagnosing that was the work; adding the index it was missing took minutes.
- **Slow-query alerting:** any query over 1s is logged and posted to the team's Discord channel, so a regression reaches the developers early instead of reaching the client.
- **Found CPU/RAM contention** between batch jobs and the app server sharing one container, on an inherited codebase with no monitoring — spotted through docker stats after users reported slowdowns. Split batch processing into a dedicated worker container with resource limits (**0.5 CPU / 512MB**) and Node.js CPU-usage self-throttling, stabilising the app server.
- **Built the cross-system sync** between the PHP estate and the ERP: B2B writes each day's business data into a shared sync database, then **12 scheduled batch jobs and 3 RabbitMQ consumers** pull it across and recompute the ERP summaries.
- **Designed and built the MES Billing module** (Laravel): multi-case billing calculation, PDF invoice generation re-mapped on edit and re-download, multi-recipient email delivery, billing list, and role-based permissions.
- Implemented real-time inventory updates with Redis, S3 and Laravel Reverb; wrote unit and feature tests for new work; ran Laravel Telescope on B2B/MES.

Tech Stack: NestJS, TypeORM, Vue 3, TypeScript, Tailwind, MySQL, PostgreSQL, RabbitMQ, Redis, Docker, Nginx, Vitest, Laravel, S3, Linux

SOTATEK COMPANY — Backend Developer

Mar 2025 – Aug 2025 | Japanese client | 16-member team, 5-person backend team

- Joined a hospital management system **from inception**, deciding the technical solutions and the initial architecture and technology stack for the **5-person backend team**.
- **Redesigned the team's Git workflow and release process**: replaced ad-hoc “dev-environment” releases with feature branches cut from main and sprint-scoped planning, so QC could plan test cases per sprint and nothing reached production without clearing QC first.
- Introduced a **peer-review checklist** and later an **AI-assisted PR review** step, cutting review overhead on out-of-scope changes. **Mentored two developers** through their onboarding onto the project.
- **Found the slow path in appointment booking**: the availability lookup across doctor schedules, shifts and existing appointments had no composite index to land on. Added one and restructured the query around how the screen actually loads its data, rather than leaving it as a single join. Also implemented OAuth authentication via Laravel Passport.

Tech Stack: Lumen, Laravel, MySQL, Redis, Docker, Nginx, S3, OAuth2/Laravel Passport, Firebase Cloud Messaging, Swagger, Linux

KOZOCOM COMPANY — Backend Developer (Middle-level)

Feb 2024 – Mar 2025 | Japanese client | 16-member team

- Built and maintained backend features for a **sales and warehouse / inventory management platform** on Laravel and PostgreSQL, working independently inside a Scrum team under clean-code and clean-architecture practices.
- **Legacy-to-new-system data migration**: CSV import of roughly **2–3 million records per run**, written as chunked insert() batches of **50 rows per worker** rather than one model create per row. At that volume the question is not speed, it is whether the job finishes before PHP runs out of memory.
- **Cut the read-path cost of the inventory list**: eager-loaded the relations it had been resolving one row at a time (a textbook N+1), and moved the list off paginate() so it stopped paying for a COUNT over the whole table on every request.
- Designed the database schemas and business logic for the purchasing, selling and store-management modules. The platform sells as well as stores: **each sale's data flows back into the warehouse, where the warehouse costs are calculated from it**.
- An **AWS SQS queue handled batched data sync** across the platform, with PDF export and fax delivery through a third-party API. Added feature and unit tests, wrote the module documentation, reviewed code and supported junior developers.

Tech Stack: Laravel, PostgreSQL, AWS SQS, Swagger, Redis, Pusher, Fax API, Nginx, Linux

FLYDINO TECHNOLOGY COMPANY — Junior Backend Developer

Apr 2023 – Feb 2024 | Japanese & Vietnamese clients | 8–18 member teams

- Independently delivered backend (Laravel) across **three client platforms**, and the frontend on two of them using Vue 3, TypeScript and Tailwind, designing the database structures for the new features.
- **Integrated ChatGPT-4** into a skin-analysis flow: an uploaded facial photo returns a skin-condition assessment that feeds a personalised product-recommendation step.
- Integrated **Keycloak SSO**, SNS login and S3/FTP storage on the recruitment platform (18-member team); reviewed and supported junior developers' code.

Tech Stack: Laravel, Vue 3, TypeScript, Tailwind, MySQL, Swagger, S3, Keycloak, Google Maps API, Nginx

NINE PLUS SOFTWARE DEVELOPMENT COMPANY — Backend Developer (Fresher → Junior)

Oct 2021 – Apr 2023 | Japanese client | 10-member team

- **Promoted from Fresher to Junior** over 18 months, moving from assigned tasks to **owning features end to end** — database schema, business logic and API — with a senior reviewing the work before it merged.
- Implemented features for an **HR management system**: designed database schemas and business logic for the recruitment and HR modules, and later for an e-commerce platform on **Stripe** — **webhooks with signature verification and idempotent handling of retried events**, the processing queued so the endpoint answered immediately, plus refund and failure paths and a reconciliation job for orders left pending.

Tech Stack: Laravel, MySQL, S3, Stripe API, Docker, Nginx

SELECTED PROJECTS

B2B ERP & MES Platform

Aug 2025 – Present | Japanese client | ERP service owned solo

- Enterprise ERP and MES covering **accounting, CRM, HR, billing, orders, shipping and inventory**. The ERP service is mine end to end; the B2B and MES systems it aggregates from are PHP, and I contribute there too. Daily sync across the two estates keeps the reporting layer current.

Tech Stack: NestJS, TypeORM, Vue 3, TypeScript, Tailwind, MySQL, PostgreSQL, RabbitMQ, Redis, Docker, Nginx, Vitest

Hospital Management System

Mar 2025 – Aug 2025 | Japanese client | 16-member team, 5-person backend | built from inception

- Appointment booking against doctor schedules and shifts, plus course, shift and contract management. OAuth via **Laravel Passport**, and push notifications. I decided the architecture and technology stack for the backend team at project start.

Tech Stack: Lumen, Laravel, MySQL, Redis, Laravel Passport, Firebase Cloud Messaging, Swagger, Docker, Nginx

Warehouse & Inventory Platform

Feb 2024 – Mar 2025 | Japanese client | 16-member team

- Sales and warehouse in one platform, replacing a legacy system: purchasing, selling and store management, with **each sale's data feeding back into the warehouse to calculate the warehouse costs**. Includes the **CSV migration that carried 2–3 million records per run** across from the legacy system, an AWS SQS queue handling batched data sync platform-wide, plus PDF export and fax delivery through a third-party API.

Tech Stack: Laravel, PostgreSQL, AWS SQS, Redis, Swagger, Pusher, Fax API, Nginx, Linux

AI Skin Analysis & Booking Platform

Dec 2023 – Feb 2024 | Japanese client | 8-member team

- **ChatGPT-4 reads an uploaded facial photo**, returns a skin-condition assessment, and feeds a personalised product-recommendation flow. Built the service-booking, store-management and authentication/authorization features end to end.

Tech Stack: Laravel (MVC), OpenAI API, JavaScript, Tailwind, MySQL, Nginx

Project Management & Recruitment Platform

Jun 2023 – Dec 2023 | Japanese client | 18-member team

- Contract, project and task modules with role-based authorization workflows. Integrated **Keycloak SSO**, SNS authentication and storage across S3 and FTP for enterprise-grade access control and file handling.

Tech Stack: Laravel (RESTful API), Vue 3, TypeScript, Tailwind, MySQL, Swagger, S3, Keycloak, Nginx

Driver Booking Platform

Apr 2023 – Jun 2023 | Vietnamese client | 8-member team

- Real-time distance and fare calculation between pickup and destination via the **Google Maps API**, plus driver booking, ratings, and driver/user management.

Tech Stack: Laravel (RESTful API), Vue 3, TypeScript, Tailwind, MySQL, Google Maps API, Nginx

E-commerce, Payments & Review

Mar 2022 – Jan 2023 | Japanese client

- **Online retail** — product, order and order-item management, CSV import/export, and online payment. Oct 2022 – Jan 2023, 9 members.
- **Store product showcase** — a platform for stores to introduce their products, with posts, review moderation, search and filtering, and ratings. Aug – Oct 2022, 7 members.
- **Multi-seller storefronts** — seller registration, product, order, order-item and store management, and the payment and shipping workflow. Mar – Aug 2022, 12–20 members.
- Integrated **Stripe** checkout on the storefronts and the online-retail platform, SNS login on the storefronts and the product-showcase platform, and Redis caching the catalog reads.

Tech Stack: Laravel (RESTful API & MVC), MySQL, Stripe API, Redis, Docker, Nginx, S3, JavaScript, Bootstrap, Pusher, Swagger

PERSONAL PROJECTS

MIA-POS — Self-funded Side Project (in testing)

Ongoing | Preview build: mia-store-remote.pages.dev

- Self-directed and self-funded POS / online-store platform, structured into domain-oriented services behind a **Caddy gateway** and split by language: **Laravel** for auth, admin and payments; **Go** for stores, inventory and the log worker; **Python** for the AI features; three **Vue 3 apps built with TypeScript** in a Bun workspace.
- The services still share one database — a modular monolith, a deliberate trade-off at this stage. **CI/CD via GitHub Actions** for automated testing and deployment.

Tech Stack: Laravel, Gin, FastAPI, PostgreSQL, Vue 3, TypeScript, Tailwind, Bun, Caddy, Cloudflare Pages, GitHub Actions, Soketi, RabbitMQ, PgBouncer

EDUCATION & LANGUAGES

FPT Polytechnic — Software Application | Jul 2019 – Dec 2021

Languages: Vietnamese — native · English — reading and writing (technical docs and specifications).